

Contribution Journal

Course / Lab: ITAI 1371 — Module 3: Machine Learning Workflow and Types of Learning

Group: ML_1371_14274 4

Student: Shareefah Adeola

My Individual Contribution

I completed my Module 3 work independently and worked through the machine learning notebook step by step. I opened the notebook in Google Colab, connected the runtime, executed the required cells, and reviewed the resulting outputs.

Data Exploration and Analysis

I examined the Wine dataset and documented the key results: 178 samples, 13 features, three classes, and no missing values. I reviewed the class distribution and feature correlations shown in the visualizations. This helped me understand why data exploration is an important part of the machine learning workflow.

Model Comparison

I reviewed the Logistic Regression and Decision Tree results and compared their accuracy. Logistic Regression achieved 88.9% accuracy, while the Decision Tree achieved 83.3%. I identified Logistic Regression as the better-performing model and explained why I believed it performed better.

Feature Experiment

I completed the feature-selection experiment using alcohol, color intensity, and proline. The selected-feature model achieved 83.3% accuracy compared with the original model's 88.9%. I documented this finding and recognized that additional features may provide useful information for classification.

Assessment and Reflection

I completed the learning-type assessment and received 5/5, or 100%. I also completed the reflection and analysis questions, including the best-performing model, possible reasons for its performance, next steps for improvement, and a real-world application in banking and finance.

Further Investigation

I added questions for further investigation about using all 13 wine features and comparing other algorithms such as Random Forest and Support Vector Machine with Logistic Regression. I also developed answers that explain what could be investigated further without claiming results that were not actually tested.

Documentation and Submission Preparation

I documented my observations in the notebook, reviewed the completed work, downloaded the completed notebook from Google Colab, and uploaded it to my GitHub repository. I also reviewed the assignment requirements and identified the need for the lab PDF, reflection journal, and contribution journal with the required naming convention.

Closing Statement

My contribution consisted of completing the required analysis, model comparison, feature experiment, assessment, reflection, further-investigation questions, and documentation. The process helped me better understand the complete machine learning workflow and how the concepts can apply to practical problems in banking and finance.